



Smart Greenhouse Monitoring And Automation System

- Using Arduino Mega, ESP8266 And Blynk Iot -

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Project Information,

- *Project Type - IoT and Embedded Systems*
- *Platform - Arduino Mega 2560*
- *Communication - ESP8266 NodeMCU Wi-Fi Module*
- *Mobile App - Blynk IoT*
- *Version - V1.0*

Main Features,

- *Password Protected Door Lock*
- *Automatic Irrigation System*
- *Temperature Controlled Ventilation*
- *Motion Activated Lighting*
- *Remote Monitoring using Blynk*
- *Real-Time Sensor Monitoring*

References,

- *Arduino Official Documentation - <https://www.arduino.cc>*
- *ESP8266 Documentation - <https://www.espressif.com>*
- *Blynk IoT Documentation - <https://docs.blynk.io>*
- *DHT11 Sensor Datasheet*
- *Soil Moisture Sensor Documentation*

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Smart Greenhouse Monitoring And Automation System

Project Overview

The Smart Greenhouse Monitoring and Automation System is an IoT-based project designed to automate environmental control inside a greenhouse. The system continuously monitors temperature, soil moisture, and motion detection, and automatically performs actions such as watering plants, controlling ventilation, managing lighting, and securing greenhouse access through a password-protected door lock.

The project uses an Arduino Mega 2560 as the main controller and an ESP8266 NodeMCU module for cloud connectivity through the Blynk IoT platform. Users can monitor sensor readings and manually control devices from a mobile application.

To make the system portable and independent from external power sources, the Arduino Mega, sensors, relays, servo motor, water pump, cooling fan, buzzer, and lighting system are powered by a **7.4V rechargeable battery pack** consisting of **two 3.7V 18650 lithium-ion batteries connected in series**. The ESP8266 NodeMCU is powered separately through its USB port using a USB power source such as a computer USB connection or a portable power bank.

Objectives

- Monitor greenhouse environmental conditions in real-time.
- Automatically irrigate plants when soil moisture is low.
- Control ventilation using temperature monitoring.
- Automatically manage lighting based on motion detection.
- Provide secure access using a keypad-controlled door lock.
- Enable remote monitoring and manual control through a mobile application.

Features

1. Smart Door Lock System

- 4x4 Membrane Keypad for password entry.
- Servo motor controls the greenhouse door.
- Password-protected access.
- Buzzer alarm activates when an incorrect password is entered.

2. Soil Moisture Monitoring and Irrigation

- Soil moisture sensor continuously monitors soil conditions.
- Water pump automatically activates when the soil becomes dry.
- Prevents overwatering and improves water efficiency.

3. Temperature Monitoring and Ventilation

- Two DHT11 temperature sensors monitor greenhouse temperature.
- Cooling fan automatically activates when temperature exceeds the threshold.
- Maintains a suitable environment for plant growth.

4. Motion Detection and Lighting System

- Two PIR motion sensors detect human movement.
- LED lighting automatically turns on when motion is detected.
- Lighting turns off automatically after a predefined period.

5. IoT Monitoring and Control

- ESP8266 NodeMCU provides Wi-Fi connectivity.
- Blynk mobile application displays sensor readings in real-time.
- Users can manually control:
 - Water Pump
 - Cooling Fan
 - Lighting System

6. Portable Power System

- Arduino Mega and peripheral devices are powered using two 18650 Li-ion batteries.
- Each battery has a capacity of 1800mAh.
- ESP8266 NodeMCU is powered through USB.
- A power bank can be used to power the ESP8266 for portable operation.
- Enables operation in locations without direct access to mains electricity.

Hardware Components

<i>Component</i>	<i>Quantity</i>
Arduino Mega 2560	1
ESP8266 Nodemcu	1
DHT11 Temperature Sensor	2
Soil Moisture Sensor	1
PIR Motion Sensor	2
4x4 Membrane Keypad	1
Servo Motor	1
Relay Module	2
Water Pump	1
Cooling Fan	1
LED Light	1
Buzzer	1
18650 Li-Ion Battery (3.7v)	2
Battery Holder (2 Cells)	1
USB Power Bank (Optional)	1
Jumper Wires	Several
Breadboard/Pcb	1

Power Supply

- The greenhouse system uses a dual power-source arrangement.
 1. Arduino Mega Power Supply
 - The Arduino Mega and all connected sensors and actuators are powered using a rechargeable battery pack made from **two 3.7V 18650 lithium-ion batteries connected in series**, providing approximately **7.4V**.
 2. ESP8266 Power Supply
 - The ESP8266 NodeMCU is powered separately through its USB port. A USB power source such as a laptop USB port, USB adapter, or portable power bank can be used.

Battery Specifications

<i>Parameter</i>	<i>Value</i>
Battery Type	18650 Lithium-Ion
Number Of Batteries	2
Voltage Per Cell	3.7v
Capacity Per Cell	1800mah

Advantages

- Portable operation without a fixed power source.
- Rechargeable and reusable batteries.
- Separate power supply improves ESP8266 stability.
- Power bank support allows flexible deployment.
- Suitable for greenhouse monitoring in remote locations.

Software and Technologies

- Arduino IDE
- C/C++ Programming
- Blynk IoT Platform
- ESP8266 Wi-Fi Module
- Serial Communication
- Embedded Systems
- IoT Monitoring

System Architecture

1. Arduino Mega Responsibilities
 - Read all sensor values.
 - Control water pump, cooling fan, and lighting.
 - Handle keypad password authentication.
 - Control servo door lock.
 - Send sensor data to ESP8266.
2. ESP8266 Responsibilities
 - Connect to Wi-Fi network.
 - Send sensor data to Blynk Cloud.
 - Receive commands from Blynk application.
 - Forward control commands to Arduino Mega.

Blynk Dashboard

- The mobile dashboard includes:
 1. Monitoring
 - Temperature Sensor 01
 - Temperature Sensor 02
 - Soil Moisture Level
 - Motion Detection Status
 2. Manual Controls
 - Fan Control Switch
 - Pump Control Switch
 - Light Control Switch

Working Principle

1. The Arduino Mega continuously reads data from all sensors.
2. Soil moisture is monitored to determine irrigation requirements.
3. Temperature values are analyzed to control ventilation.
4. PIR sensors detect movement and control greenhouse lighting.
5. Users unlock the greenhouse door using a predefined password.
6. Sensor data is transmitted to the ESP8266.
7. ESP8266 uploads data to the Blynk cloud platform.
8. Users monitor the greenhouse remotely using the Blynk mobile application.
9. Users can manually control devices from anywhere through the internet.
10. The Arduino Mega and connected devices operate using rechargeable 18650 batteries, while the ESP8266 is powered through USB or a power bank.

Results

The developed system successfully:

- Automated greenhouse irrigation.
- Automated temperature-based ventilation.
- Implemented motion-based lighting control.
- Provided secure password-protected access.
- Enabled real-time remote monitoring.
- Enabled manual remote device control using Blynk.
- Operated reliably using rechargeable batteries and USB-powered Wi-Fi connectivity.

The project demonstrated reliable performance and effective greenhouse automation using low-cost IoT technologies.

Future Improvements

- Add humidity monitoring and control.
- Implement automatic fertilization system.
- Integrate weather forecasting APIs.
- Add camera-based remote monitoring.
- Store sensor data in cloud databases for analysis.
- Implement mobile notifications and alerts.
- Integrate solar charging for extended battery life and sustainable operation.

Conclusion

The Smart Greenhouse Monitoring and Automation System successfully integrates embedded systems, sensors, automation, portable power solutions, and IoT technologies to create an intelligent greenhouse environment. The system improves plant care efficiency, reduces manual intervention, conserves resources, and enables remote monitoring and control through the Blynk platform. The use of rechargeable 18650 batteries for the Arduino Mega system and USB or power bank power for the ESP8266 provides flexibility, portability, and reliable operation in locations without a continuous external power supply.